EXPERIMENT 6

6	An online food delivery platform enables customers to browse restaurants, view menus, place food orders, make online payments, and track delivery status. Restaurants use the system to manage their menus and update order status, while delivery agents are responsible for accepting delivery requests and updating delivery progress. Design a Class Diagram, Use case Diagram and Activity Diagram for the Online Food Delivery System showing the relevant classes, their attributes, methods, and relationships.	Umbrello	CO1

Objective

To design UML diagrams (Class, Use Case, and Activity) for an Online Food Delivery System using Umbrello, representing system functionality, structure, and workflow.(mythresree,losane)

System Description (Short)

The Online Food Delivery System allows:

- Customers → browse restaurants, order food, make payments, track delivery
- Restaurants → manage menu and update order status
- Delivery Agents → accept orders and update delivery progress

🔹 PART 1: CLASS DIAGRAM

Step 1: Open Umbrello

1. Open **Umbrello UML Modeller**
2. Click **File → New**

Step 2: Select Class Diagram

1. In left panel → Right click **Logical View**
2. Select **New Diagram → Class Diagram**
3. Rename as:

👉FoodDelivery_ClassDiagram

Step 3: Add Classes

Click **Class icon** and create:

1. Customer
2. Restaurant
3. Menu
4. FoodItem
5. Order
6. Payment
7. DeliveryAgent

Step 4: Add Attributes & Methods

Customer

- customerId
- name
- address
- phone

Methods:

- browseRestaurants()
- placeOrder()
- makePayment() □ trackOrder()

Restaurant

- restaurantId
- name
- location

Methods:

- updateMenu() □ acceptOrder()

Menu

- menuId

Methods:

- showItems()

FoodItem

- itemId

- itemName □ price

Order

- orderId
- status Methods:
- updateStatus()

Payment

- paymentId
- amount
- paymentStatus Methods:
- processPayment()

DeliveryAgent

- agentId
- name

Methods:

- acceptDelivery()
- updateDeliveryStatus()

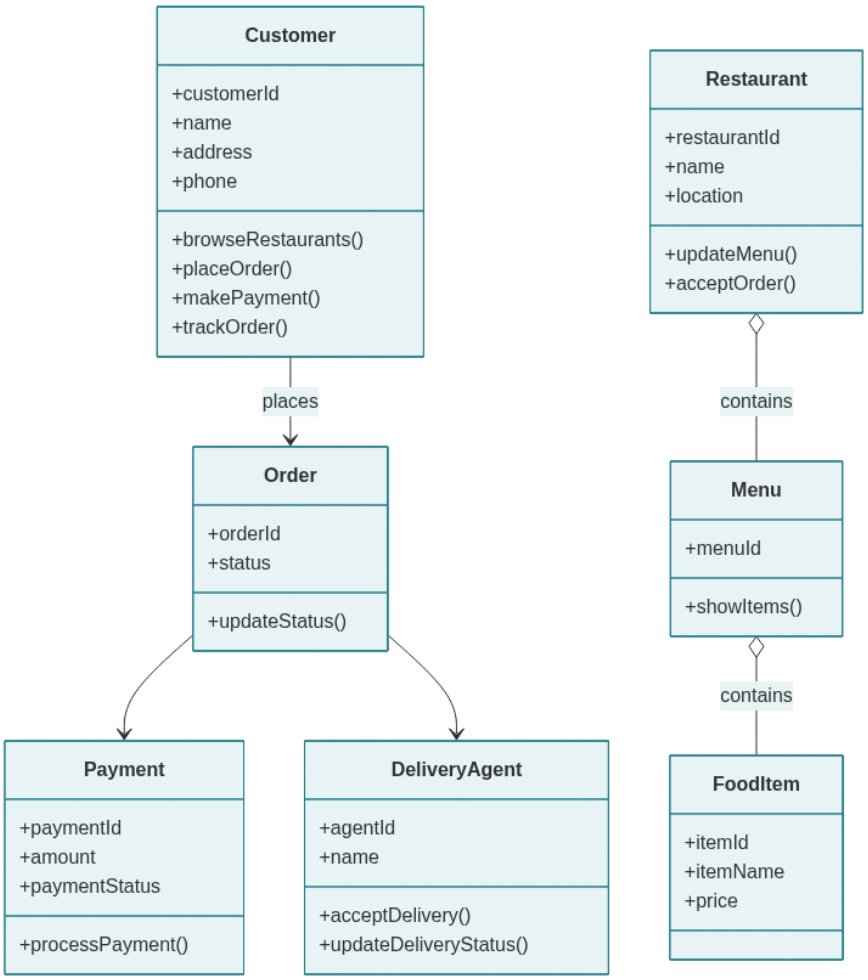
Step 5: Draw Relationships

Use **Association / Aggregation lines**:

- Customer → Order
- Restaurant → Menu
- Menu → FoodItem
- Order → Payment
- Order → DeliveryAgent

Step 6: Save Diagram

File → Save as FoodDelivery_ClassDiagram.uxf



◆ PART 2: USE CASE DIAGRAM

Step 1: Create Use Case Diagram

Right click **Use Case View → New Diagram**
Rename as: FoodDelivery_UseCaseDiagram

Step 2: Add Actors

Add:

- Customer
- Restaurant
- Delivery Agent

Step 3: Add Use Cases

Customer Use Cases

- Browse Restaurants
- View Menu
- Place Order
- Make Payment
- Track Delivery

Restaurant Use Cases

- Manage Menu
- Update Order Status

Delivery Agent Use Cases

- Accept Delivery
- Update Delivery Status

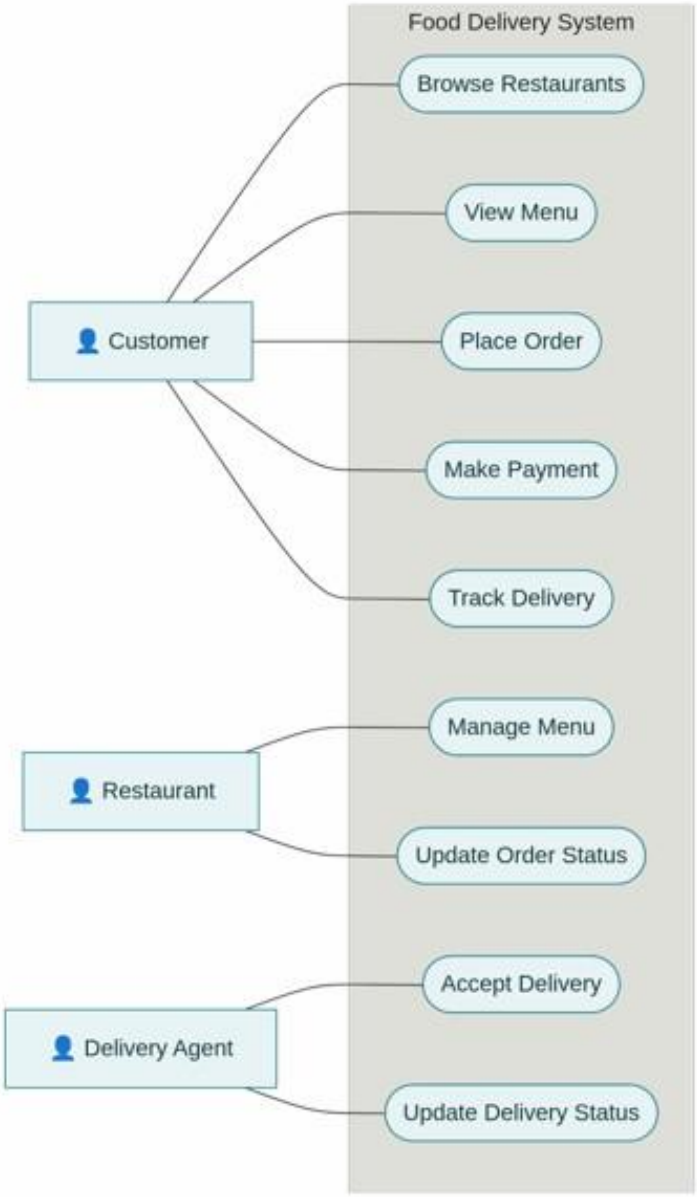
Step 4: Connect Actors to Use Cases

Use **Association line**:

- Customer → all customer use cases
- Restaurant → manage menu, update order Delivery Agent → delivery tasks

Step 5: Save Diagram

Save as: FoodDelivery_UseCaseDiagram.uxf



- **PART 3: ACTIVITY DIAGRAM**

Step 1: Create Activity Diagram

Right click **Logical View → New Diagram → Activity Diagram**

Rename as: FoodDelivery_ActivityDiagram

Step 2: Add Activities in Sequence

1. Start
2. Browse Restaurants
3. View Menu
4. Select Food
5. Place Order
6. Make Payment
7. Order Confirmed
8. Restaurant Prepares Food
9. Delivery Agent Picks Order
10. Deliver Food
11. Order Completed
12. Stop

Step 3: Connect Flow

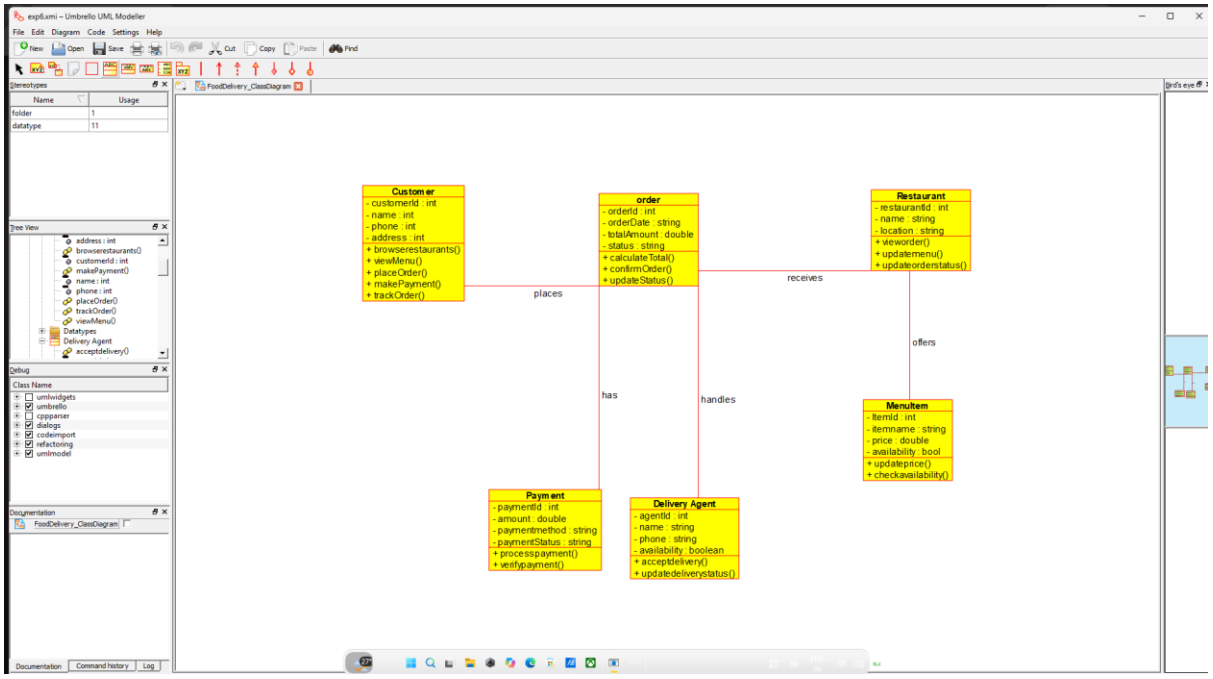
Use **Control Flow arrows** to connect each activity in order.

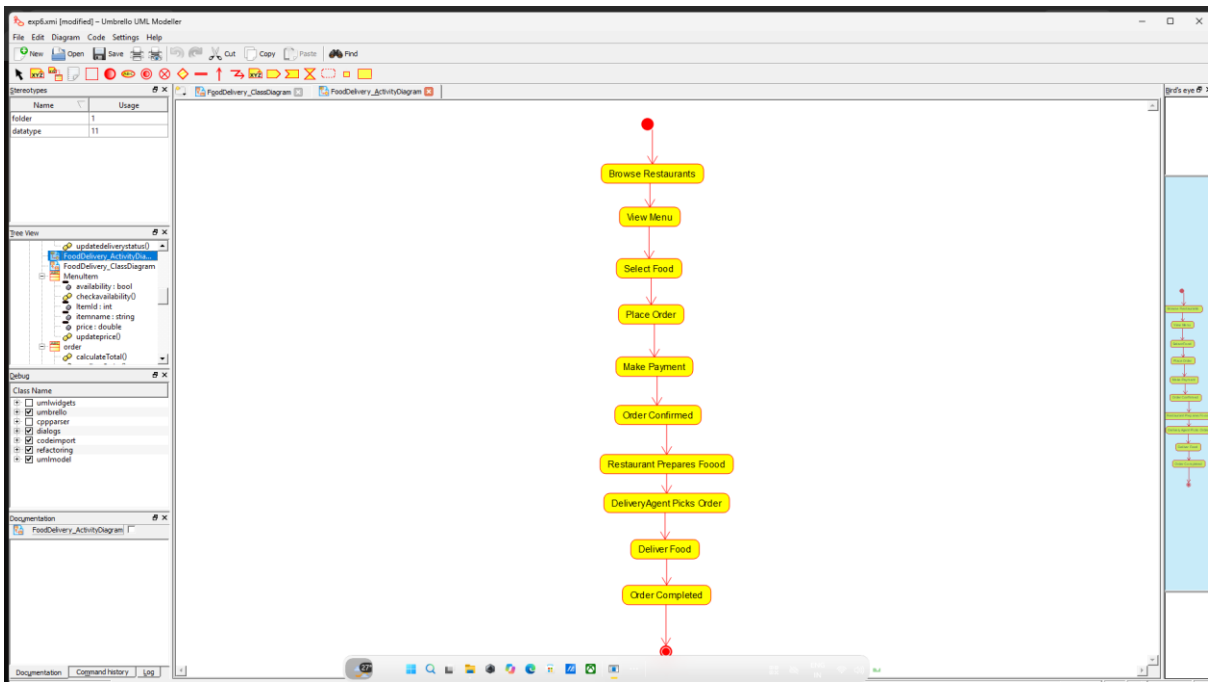
Step 4: Save Diagram

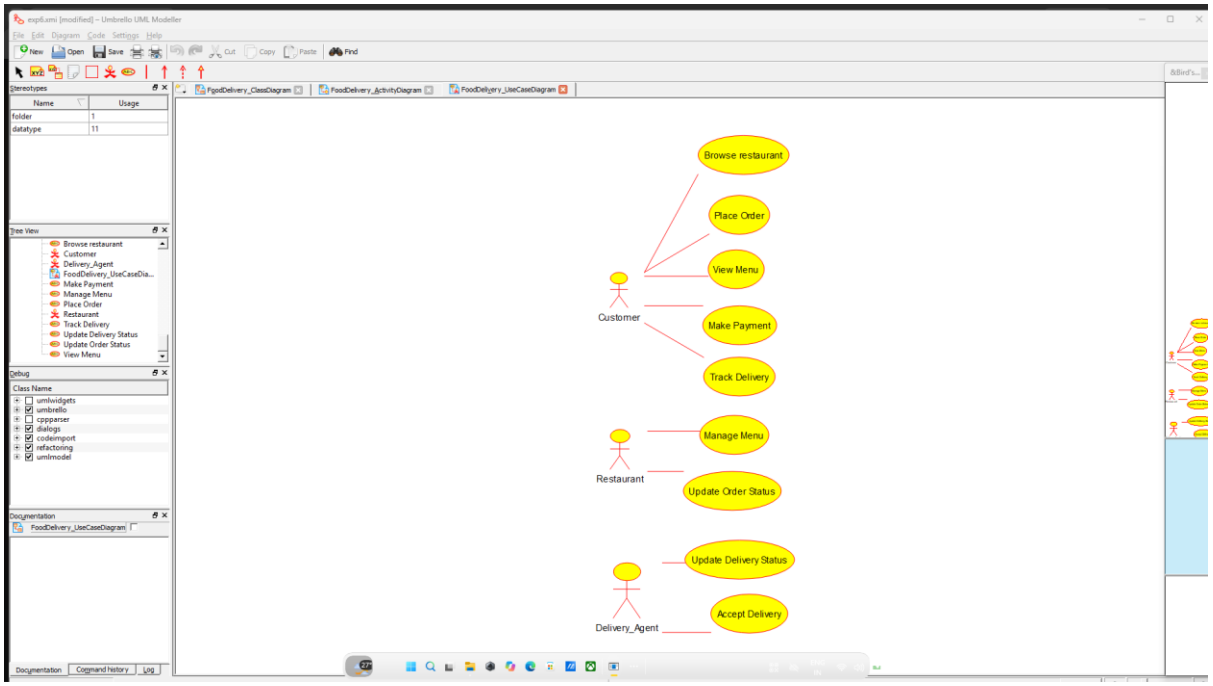
Save as: FoodDelivery_ActivityDiagram.uxf

Result

Thus, Class Diagram, Use Case Diagram, and Activity Diagram for the Online Food Delivery System were successfully designed using Umbrello UML Modeller.


exp6.xmi – Umbrello UML Modeller
File Edit Diagram Code Settings Help
FoodDelivery_ClassDiagram
Customer
- customerId : int
- name : int
- phone : int
- address : int
+ browserestaurants()
+ viewMenu()
+ placeOrder()
+ makePayment()
+ trackOrder()
order
- orderId : int
- orderDate : string
- totalAmount : double
- status : string
+ calculateTotal()
+ confirmOrder()
+ updateStatus()
Restaurant
- restaurantId : int
- name : string
- location : string
+ vieworder()
+ updatemenu()
+ updateorderstatus()
places
receives
offers
has
handles
MenuItem
- ItemId : int
- itemname : string
- price : double
- availability : bool
+ updateprice()
+ checkavailability()
Payment
- paymentId : int
- amount : double
- paymentmethod : string
- paymentStatus : string
+ processpayment()
+ verifypayment()
Delivery Agent
- agentId : int
- name : string
- phone : string
- availability_boolean
+ acceptdelivery()
+ updatedeliverystatus()


exp6.xmi [modified] – Umbrello UML Modeller
File Edit Diagram Code Settings Help
FoodDelivery_ClassDiagram
FoodDelivery_ActivityDiagram
Browse Restaurants
View Menu
Select Food
Place Order
Make Payment
Order Confirmed
Restaurant Prepares Foood
DeliveryAgent Picks Order
Deliver Food
Order Completed


Browse restaurant
Place Order
View Menu
Customer
Make Payment
Track Delivery
Manage Menu
Restaurant
Update Order Status
Update Delivery Status
Accept Delivery
Delivery_Agent